Vectors and Matrices Assignment

28/02/2026

Question 1

Show that

$$a \cdot (b \times c)$$

is equal to the determinant whose rows are respectively the components of the vectors a, b, c.

Hence, prove that this gives the volume of the parallelepiped formed by the vectors a, b, c.

$$a \cdot (b \times c) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

—

Question 2

Use the formula in Exercise 1 to calculate the volume of the tetrahedron having vertices:

$$(0,0,0), \quad (0,-1,2), \quad (0,1,-1), \quad (1,2,1)$$

Recall that:

$$\text{Volume of tetrahedron} = \frac{1}{6}\,|a \cdot (b \times c)|$$

—

Question 3

Show that if three origin vectors lie in the same plane, the determinant having the three vectors as its rows has value zero.

—

Question 4: Cross Product

Find $A \times B$ if:

(a)

$$A = \mathbf{i} - 2\mathbf{j} + \mathbf{k}, \quad B = 2\mathbf{i} - \mathbf{j} - \mathbf{k}$$

(b)

$$A = 2\mathbf{i} - 3\mathbf{k}, \quad B = \mathbf{i} + \mathbf{j} - \mathbf{k}$$

Vectors and Matrices

1D-2

Find the area of the triangle in space having its vertices at the points

$$P(2, 0, 1), \quad Q(3, 1, 0), \quad R(-1, 1, -1).$$

—

1D-3

Two vectors $\mathbf{i}'$ and $\mathbf{j}'$ of a right-handed coordinate system are to have the directions respectively of the vectors

$$A = 2\mathbf{i} - \mathbf{j}$$

and

$$B = \mathbf{i} + 2\mathbf{j} + \mathbf{k}.$$

Find all the three vectors $\mathbf{i}', \mathbf{j}', \mathbf{k}'$.

—

1D-4

Verify that the cross product $\times$ does not in general satisfy the associative law, by showing that for the particular vectors $\mathbf{i}, \mathbf{j}, \mathbf{k}$ we have

$$(\mathbf{i} \times \mathbf{j}) \times \mathbf{k} \neq \mathbf{i} \times (\mathbf{j} \times \mathbf{k}).$$

—

1D-5

What can you conclude about A and B if:

(a) $|A \times B| = |A||B|$?

(b) $|A \times B| = A \cdot B$?

—

1D-6

Take three faces of a unit cube having a common vertex P. Each face has a diagonal ending at P.

What is the volume of the parallelepiped having these three diagonals as coterminous edges?

—

1D-7

Find the volume of the tetrahedron having vertices at the four points

$$P(1,0,1), \quad Q(-1,1,2), \quad R(0,0,2), \quad S(3,1,-1).$$

Hint:

$$\text{Volume of tetrahedron} = \frac{1}{6}\left|a \cdot (b \times c)\right|.$$

Hint

Volume of tetrahedron $= \dfrac{1}{6}$ (Volume of parallelepiped with the same three coterminous edges)

—

1D-8

Prove that

$$A \cdot (B \times C) = (A \times B) \cdot C$$

by using the determinantal formula for the scalar triple product and the algebraic laws of determinants.

—

1D-9

Show that the area of a triangle in the xy-plane having vertices

$$(x_1, y_1), \quad (x_2, y_2), \quad (x_3, y_3)$$

is given by the determinant

$$\frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}.$$

Do this:

(a) By relating the area of the triangle to the volume of a certain parallelepiped.

(b) By using the laws of determinants (p. 170 of the notes) to relate this determinant to the 2×2 determinant normally used to calculate the area.

IE 1: Equation of Lines and Planes

Find the equation of the following planes:

1. Through $(2, 0, -1)$ and perpendicular to $\mathbf{i} + 2\mathbf{j} - 2\mathbf{k}$.
2. Through the origin and the points $(1, 1, 0)$ and $(2, -1, 3)$.
3. Through $(1, 0, 1)$, $(2, -1, 2)$ and $(-1, 3, 2)$.
4. Through the points on the x, y, and z axes where $x = a$, $y = b$, $z = c$ respectively. Give the equation in the form $Ax + By + Cz = D$.
5. Through $(1, 0, 1)$ and $(0, 1, 1)$ and parallel to $\mathbf{i} - \mathbf{j} + 2\mathbf{k}$.

Equations of Lines and Planes

IE–1: Equation of Lines and Planes

Find the equation of the following planes:

1. Through $(2, 0, -1)$ and perpendicular to $\mathbf{i} + 2\mathbf{j} - 2\mathbf{k}$.
2. Through the origin and the points $(1, 1, 0)$ and $(2, -1, 3)$.
3. Through $(1, 0, 1)$, $(2, -1, 2)$ and $(-1, 3, 2)$.
4. Through the points on the x, y, and z axes where $x = a$, $y = b$, $z = c$ respectively. Give the equation in the form $Ax + By + Cz = D$.
5. Through $(1, 0, 1)$ and $(0, 1, 1)$ and parallel to $\mathbf{i} - \mathbf{j} + 2\mathbf{k}$.

IE–2

Find the dihedral angle between the planes

$$2x - y + z = 3 \quad \text{and} \quad x + y + 2z = 1.$$

IE–3

Find, in parametric form, the equation of the line:

1. Through $(1, 0, -1)$ and parallel to $2\mathbf{i} - \mathbf{j} + 3\mathbf{k}$.
2. Through $(2, -1, -1)$ and perpendicular to the plane

$$2x - y + 2z = 3.$$

3. All lines passing through $(1, 1, 1)$ and lying in the plane

$$2x + 2y - z = 2.$$

IE–4

Where does the line through $(0, 1, 2)$ and $(2, 0, 3)$ intersect the plane

$$x + 4y + z = 4\,?$$

IE–5

The line passing through $(1, 1, -1)$ and perpendicular to the plane

$$3x + 2y - z = 3$$

intersects the plane

$$2x - y + z = 1.$$

Find the point of intersection.

IE–6

Show that the distance D from the origin to the plane

$$ax + by + cz = d$$

is given by

$$D = \frac{|d|}{\sqrt{a^2 + b^2 + c^2}}.$$

Hint: Let $\mathbf{n}$ be a unit normal to the plane and P be a point on the plane. Consider the component of $\overrightarrow{OP}$ in the direction of $\mathbf{n}$.

IE–7

Formulate a general method for finding the distance between two skew (i.e. non–intersecting) lines in space and carry it out for two non–intersecting lines lying along the diagonals of two adjacent faces of the unit cube (placed in the first octant, with one vertex at the origin).

Hint: The shortest line segment joining the two skew lines will be perpendicular to both of them (if it were not, it could be shortened).

IF–1 Matrix Algebra

Let

$$A = \begin{pmatrix} 2 & -1 & 3 \\ 1 & 0 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & -1 \\ 2 & 3 \\ -1 & 2 \end{pmatrix}, \quad C = \begin{pmatrix} 0 & 2 \\ 3 & 4 \\ 1 & 4 \end{pmatrix}.$$

Compute:

(a) $B + C, \quad B - C, \quad 2B - 3C.$

(b) $AB, \quad AC, \quad CA, \quad BC^T, \quad CB^T.$

(c) $A(B + C), \quad AB + AC, \quad (B + C)A, \quad BA + CA.$

IF–2

Let A be an arbitrary $m \times n$ matrix and let I_k be the identity matrix of size k. Verify that

$$I_m A = A \quad \text{and} \quad AI_n = A.$$

IF–3

Find all 2×2 matrices

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

such that

$$A^2 = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

IF–4 Vectors and Matrices

Show that matrix multiplication is not, in general, commutative by calculating, for each pair below, the matrices AB and BA.

(a)

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

(b)

$$A = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 1 & 2 \\ 1 & 2 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & -2 & 1 \\ 3 & -2 & 4 \\ 3 & 5 & -1 \end{pmatrix}.$$